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Unlocking Field-Wide Value: Integrated Gas Plateau Extension Strategies Across Recycle and Depletion Reservoirs

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Abstract

This paper presents a simulation-based redevelopment strategy applied across a portfolio of mature gas condensate reservoirs. The approach aims to extend production plateau, optimize condensate recovery, and delay abandonment. Dynamic reservoir models were used to evaluate five gas recycling reservoirs and one depletion driven asset. Five strategic levers were assessed: GI to GP conversion, tubing head pressure (THP) reduction, optimized blowdown scheduling, deployment of compression, and strategic infill drilling. Simulation results reveal that when applied in a phased, integrated manner, these levers yielded substantial increases in recovery efficiency and plateau life without significant capital burden. Findings demonstrate the value of multi reservoir simulation frameworks in shaping field wide optimization strategies, especially in constrained infrastructure and CAPEX limited environments.

Introduction

Gas condensate reservoirs are characterized by complex phase behavior, high pressure gas, and liquid hydrocarbon condensation as reservoir pressure falls below dew point. In late life fields, the dominant challenges include liquid banking, declining productivity, and diminished reservoir energy, all of which threaten plateau sustainability and overall recovery. This is exacerbated in mature assets with limited CAPEX flexibility and aging infrastructure. The present study addresses these challenges by employing a dynamic, simulation-based framework that integrates multiple development strategies across a portfolio of reservoirs. The goal is to identify and quantify opportunities to extend plateau and enhance liquids recovery while minimizing new investments.

The development levers explored include Gas Injection (GI) to Gas Producer (GP) conversion, THP reduction, blowdown optimization, compression strategies, and infill drilling. Each lever was modeled using history matched simulation models. Performance was evaluated using key metrics such as plateau duration, gas rate uplift, condensate recovery factor (CRF), and value to cost ratio. A strong emphasis was placed on evaluating synergies between levers, reservoir specific behavior, and system wide implications across the gas infrastructure. The paper presents technical findings, simulation workflows, and decision criteria to support optimal redevelopment planning.

To guide simulation design and ensure consistency across reservoirs, development levers were categorized into two primary streams: Depletion Gas and Gas Recycle. As illustrated in Fig. 1, the Depletion Gas branch included inline wellhead compression, infill wells for production acceleration, targeted gas injection in low-pressure depleted reservoirs, and full-field compression proposals. These strategies focused on accelerating reservoir pressure drawdown while safeguarding surface deliverability.

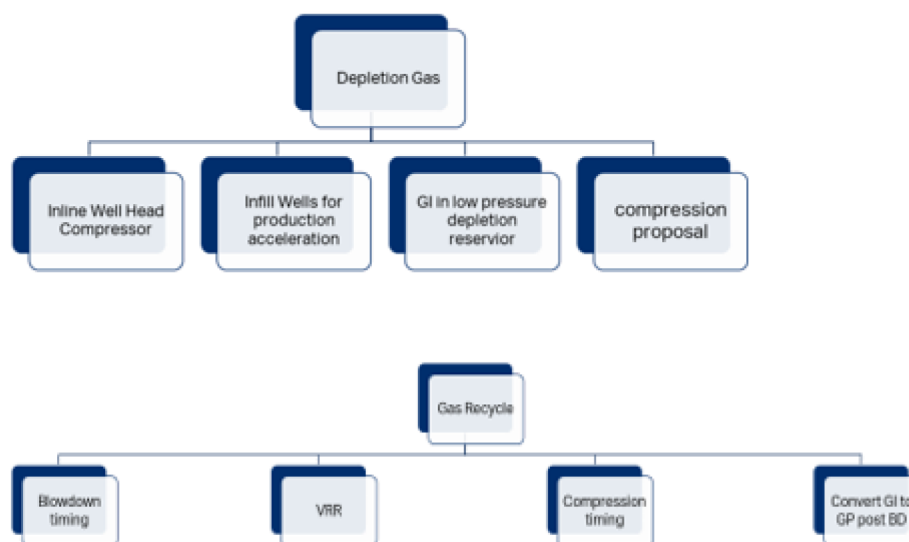


Figure 1—Strategic Classification of Development Levers

Conversely, the Gas Recycle branch aimed to enhance gas-cap utilization and defer decline through four key levers: blowdown timing, Voidage Replacement Ratio (VRR) optimization, compression timing, and post-blowdown conversion of gas injectors to gas producers. Each lever was assessed individually and in phased combinations to reflect realistic implementation pathways.

This classification enabled a structured approach to scenario design and facilitated clear performance comparison across reservoirs. It also ensured coverage of both short-term acceleration and long-term recovery strategies, while capturing interactions between subsurface development and surface constraints.

Methodology

Each reservoir in this study was represented using high-resolution, multimillion-cell compositional models constructed in the tNav simulator. For the Depletion Reservoirs, both standalone models for each individual reservoir and integrated models combined with a single surface network were developed and utilized. This modeling approach enabled detailed simulation of complex reservoir behavior, fluid interactions, and surface facility constraints across the entire portfolio.

Scenarios were designed to test phased implementation of development levers. For example, one case introduced THP reduction before infill drilling, while another reversed the sequence. GI to GP conversions were tested under three timing assumptions: early plateau, mid plateau, and late plateau. Blowdown cases were staged based on gas plant availability and field wide cumulative injection thresholds. Compression deployment cases accounted for suction pressure, discharge pressure, and compression ratio efficiencies. In total, more than 80 combinations of timing, sequence, and constraint assumptions were tested.

All simulation models employed in this study were constructed using a structured gridding approach, incorporating layered heterogeneity and regionally variable permeability and porosity distributions. Each model was built to represent the full field, including gas cap, oil rim (where applicable), and aquifer regions. The grid resolution was refined in the near wellbore regions to accurately capture pressure gradients,

drawdown effects, and potential coning behavior. Layer discretization was aligned with well log correlation data and seismic structural interpretation.

This study utilized fully compositional dynamic simulation models across six reservoirs. The models incorporate rock and fluid properties, relative permeability hysteresis, capillary pressure, and well level operational constraints. Each model was independently history matched to field performance data including gas, condensate, and water rates; bottomhole pressure; and voidage replacement ratios (VRR). Simulation forecasts covered multiple operational scenarios spanning 20+ years of production.

The primary focus was on five development levers:

1. **GI to GP Conversion:** This involved reconfiguring gas injectors into producers as pressure declined, or plateau dropped below target. Simulation inputs included changes in well control strategy, skin sensitivity, and dynamic mobility ratios.
2. **THP Reduction:** Multiple tubing head pressure targets (600 psia, 450 psia, 200 psia, and 50 psia) were tested. Surface backpressure nodes were linked to the subsurface model using integrated network modeling tools to represent compressor suction constraints.
3. **Blowdown Optimization:** Timed cessation of injection and subsequent rapid pressure drawdown scenarios were tested. The onset of blowdown was staged and sequenced across reservoirs to minimize peak gas load and surface bottlenecks.
4. **Compression Deployment:** Inline and centralized compression were introduced through well level WHP constraints and group level node reductions. Compression phasing was tested based on forecasted reservoir pressure depletion and facility readiness.
5. **Infill Drilling:** New wells were placed based on remaining gas in place (RGIP) maps. Optimized trajectories targeted bypassed pockets, preferential flow corridors, and zones of underutilized permeability. Each infill case was constrained by slot availability and surface handling capacity.

Sensitivity analyses were run for each lever using varied inputs: well PI degradation, injector skin evolution, condensate dropout timing, GOR limits, and reservoir connectivity. Performance was evaluated against a base case with continued injection and no new development. In total, over 80 scenario simulations were completed, and results were compiled into a performance matrix indexed by plateau gain, gas uplift, and condensate yield.

Results

Quantitative comparison across scenarios showed that GI to GP added an average of 6.5 years of plateau in the top two reservoirs, while adding only 1.1 years in the lowest. THP reduction at 200 psia increased cumulative gas production by an average of 8.7%, while infill drilling contributed an additional 5.2% in recoverable gas when strategically placed. Blowdown initiation timing, when delayed to a specific pressure threshold (typically ~1200 psi), protected condensate yields by ~4% on average. These findings emphasize the value of scenario-based lever screening using validated models.

Reservoir E, a structurally compartmentalized reservoir with complex faulting, responded poorly to infill drilling due to localized depletion. Pressure maps showed that most of the new wells were drilled in pressure depleted blocks disconnected from the main producing structure. This case highlighted the importance of incorporating 3D connectivity modeling before planning infill programs. Despite these challenges, coupling compression with THP reduction helped partially recover flow capacity in both reservoirs.

In addition to top performing reservoirs, two assets reservoir C and E showed limited responsiveness to several levers. In reservoir C, a highly laminated formation with low vertical permeability, GI to GP conversion delivered only 1.2 years of plateau extension. Tracer simulation indicated poor vertical sweep and limited injector connectivity, explaining the marginal benefit. Condensate recovery improved by only

0.8%, and skin evolution data suggested that conversion induced drawdown created near wellbore liquid loading.

Fig. 2 demonstrates the production response of reservoir D to gas injector (GI) to gas producer (GP) conversion executed after blowdown. The red curves represent the base case scenario without conversion, while the green curves show the performance with GI to GP conversion.

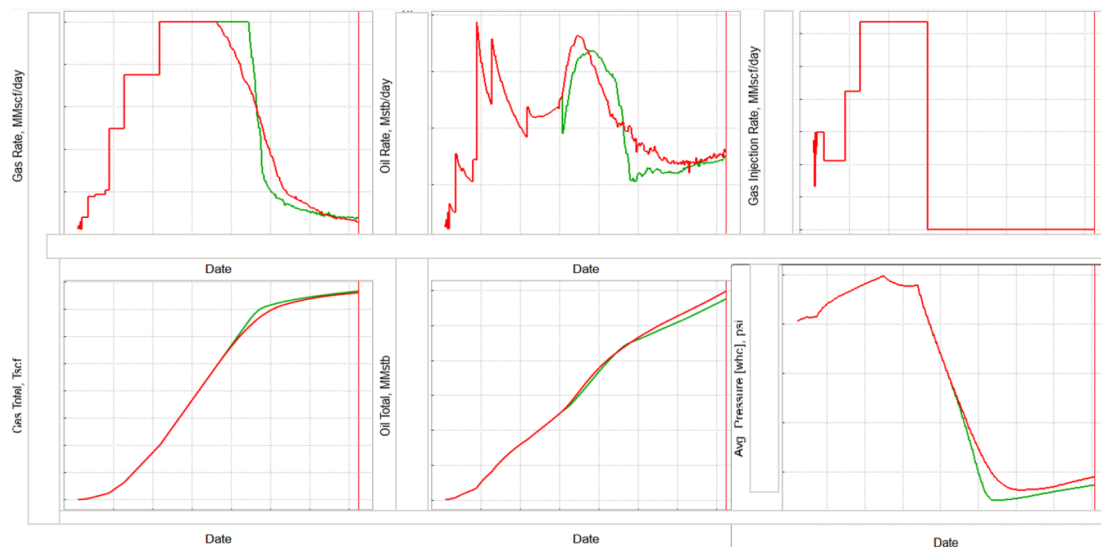


Figure 2—GI to GP Conversion Post-Blowdown for reservoir D

The gas rate plot indicates a slight but measurable uplift in post-blowdown gas production when the injector is converted. However, the gains were short-lived, reflecting the reservoir's limited connectivity and the localized nature of pressure support around the former injector. The oil rate curve shows a brief surge in condensate production, but the total condensate recovery (oil total) increased by less than 1% — confirming the marginal benefit of the conversion in this case.

The gas injection rate drops to zero in both cases as blowdown commences, but only the GI-to-GP case sustains a small production leg post-conversion. The average reservoir pressure drops more rapidly in the converted case, suggesting that drawdown near the wellbore was not effectively translated into fieldwide production benefits. This aligns with prior tracer studies that revealed limited communication between the injector and surrounding drainage volumes.

Reservoir D is characterized by compartmentalized flow units and limited vertical permeability, which reduces the effectiveness of injector repurposing. The drawdown created post-conversion did not induce meaningful pressure gradients capable of accelerating fluid movement from unswept regions. This figure highlights the importance of assessing reservoir architecture, injector placement, and flow potential when selecting candidates for GI to GP conversion.

In summary, while GI to GP conversion is a powerful tool in plateau extension, reservoir D illustrates the limitations of applying this strategy in low-connectivity systems. The technical feasibility must be grounded in robust dynamic modeling and tracer diagnostics to avoid inefficient deployments.

The simulation results revealed a differentiated impact of each development lever across the reservoir portfolio. In GI reservoirs with strong connectivity and maintained VRR, GI to GP conversion extended plateau by 6.11 years. This was attributed to high deliverability of converted wells, combined with mobilized condensate previously trapped near injectors. In reservoirs with poor communication or limited GI history, conversion provided minimal uplift and required significant skin correction.

Fig. 3 illustrates the effect of varying Voidage Replacement Ratio (VRR) through additional gas injection prior to blowdown in reservoir B. This reservoir, characterized by high-pressure gas-cap support and

moderate condensate yield, was used to study how reservoir energy optimization influences post-injection production.

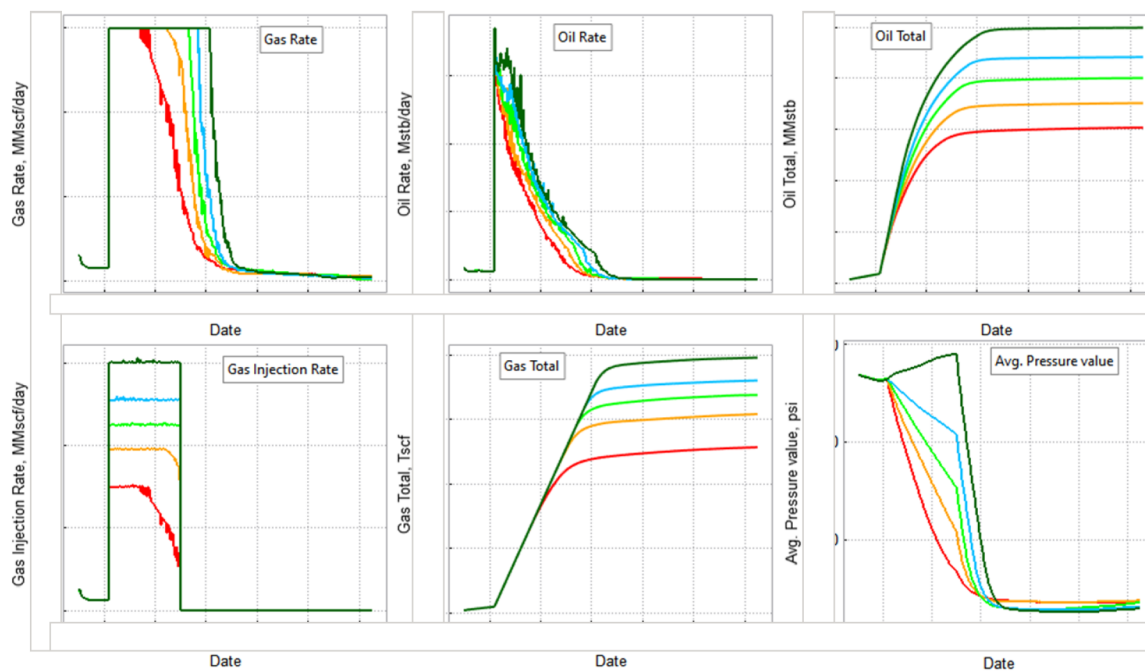


Figure 3—reservoir B: Impact of Pre-Blowdown Injection Volume (VRR Sensitivity)

In this figure, the red line represents the base case, where blowdown was initiated at a reference VRR level. Other curves represent cases with progressively higher injection volumes, resulting in increased VRR targets (up to near-unit replacement).

Key technical findings:

- **Gas Rate & Plateau Duration:** Higher pre-blowdown injection volumes resulted in prolonged gas plateau and smoother decline trends. Pressure reinforcement near the injectors enhanced gas deliverability, delaying plateau collapse by multiple years compared to the base case.
- **Oil (Condensate) Recovery:** Incremental gains in condensate recovery were observed with increasing VRR. This was attributed to improved drawdown control and reduced near-wellbore liquid dropout due to pressure support.
- **Pressure Behavior:** The average reservoir pressure plot shows clear pressure stratification across cases, with higher VRR preserving energy and delaying critical pressure drop. This behavior supported both gas and condensate flow efficiency.
- **Gas Injection Termination:** All scenarios terminated injection at the same cut-off point, allowing a clean comparison of pressure buildup effects. Reservoirs with higher voidage replacement showed a more favorable transition into blowdown, reducing sharp flow disruptions.
- **Gained Efficiency Without Conversion:** Notably, this case did not involve any GI to GP conversion. Instead, it validated the value of targeted over-injection before blowdown as a low-risk, high-reward lever for production optimization.

This figure confirms that injecting more gas before switching to depletion can significantly improve both plateau duration and hydrocarbon recovery. In high-GOR systems like reservoir B, this approach can be critical in maximizing value during the transition from injection to blowdown, especially where compression infrastructure is limited or delayed.

THP reduction resulted in measurable flowrate increases across all reservoirs. At 450 psia, production uplift averaged 18–27%. However, reducing THP to 50 psia introduced marginal gains beyond 200 psia and raised mechanical concerns in legacy completions. The most favorable cost performance outcome was achieved at 200 psia, especially when paired with compressor installation in the surface network.

Fig. 4 illustrates the impact of tubing head pressure (THP) on production performance during blowdown for a high-pressure gas-condensate reservoir (reservoir B). The simulation compares three scenarios: a base case at 450 psia (red), a reduced THP case at 200 psia (green), and a more aggressive reduction to 50 psia (blue).

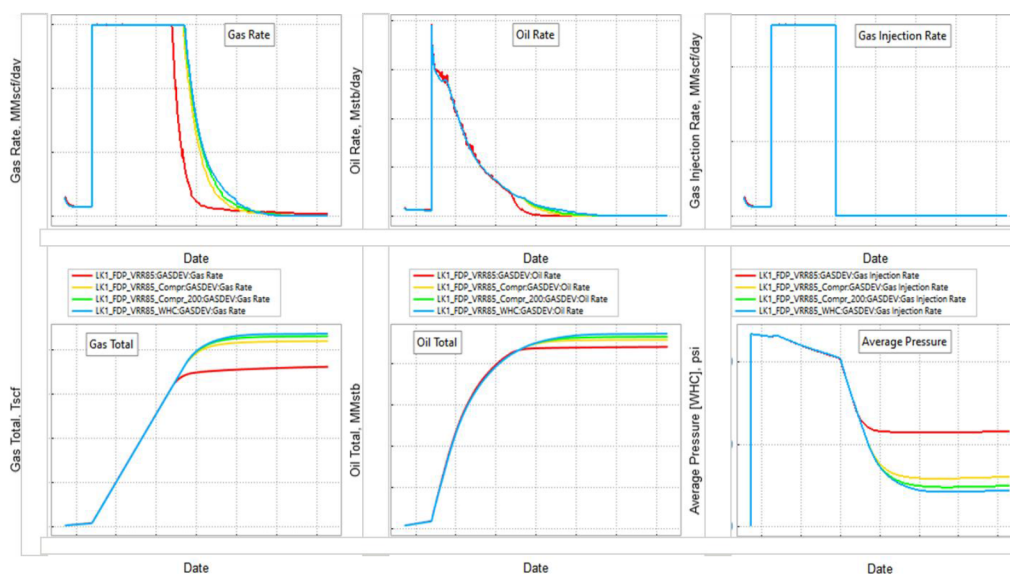


Figure 4—reservoir B: Impact of Flowing THP to production performance

Reduction of THP clearly enhances gas deliverability by minimizing wellbore backpressure. The 200 psia case exhibits a significant increase in early gas rate relative to the base case, while the 50 psia case provides a marginally higher uplift but introduces operational risks and faster depletion.

Condensate recovery, represented by oil rate and cumulative liquid yield, shows a clear trade-off. The base case (450 psia) retains higher average reservoir pressure longer, preserving condensate dropout and recovery. In contrast, the 50 psia case yields the lowest condensate volume due to rapid pressure drawdown, promoting gas liberation and liquid loss. The 200 psia scenario offers a balanced outcome—delivering higher gas rates with only moderate compromise in liquid recovery.

Reservoir pressure trends confirm the accelerated depletion at lower THPs. While this improves gas rate, it may challenge completion integrity and limit long-term recovery if not mitigated through compression.

Overall, 200 psia THP emerged as the most cost-effective option. It enhanced early gas plateau duration while maintaining reasonable condensate yield and minimizing mechanical exposure. Gains beyond 200 psia offered diminishing returns and increased operational complexity, especially in legacy completions. This finding underscores the importance of optimizing drawdown strategy based on reservoir fluid properties and surface constraints.

Blowdown optimization varies widely. In high pressure, high GOR systems, early blowdown resulted in gas rate surges but compromised condensate recovery. Conversely, delayed blowdown preserved liquid yield but prolonged gas plateau at declining rates. Optimal blowdown timing was found by balancing surface sales commitments with condensate yield targets. One reservoir demonstrated a 3.9% increase in condensate recovery by deferring blowdown by 6 years. The following figure (Fig. 5) shows the following:

Sensitivity to Blowdown Timing:

Each colored line represents a different blowdown start year. As the start year is pushed later, the gas plateau duration extends significantly, delaying decline and maintaining sales rates longer.

- Plateau Extension Benefit:

Deferring blowdown allowed the reservoir to deliver up to 6 additional years of production at plateau rates. This is attributed to maintaining higher reservoir pressure and improved liquid carryover.

- Condensate Yield Uplift:

The delayed BD case also enhanced condensate recovery reinforcing the importance of timing in depletion strategy.

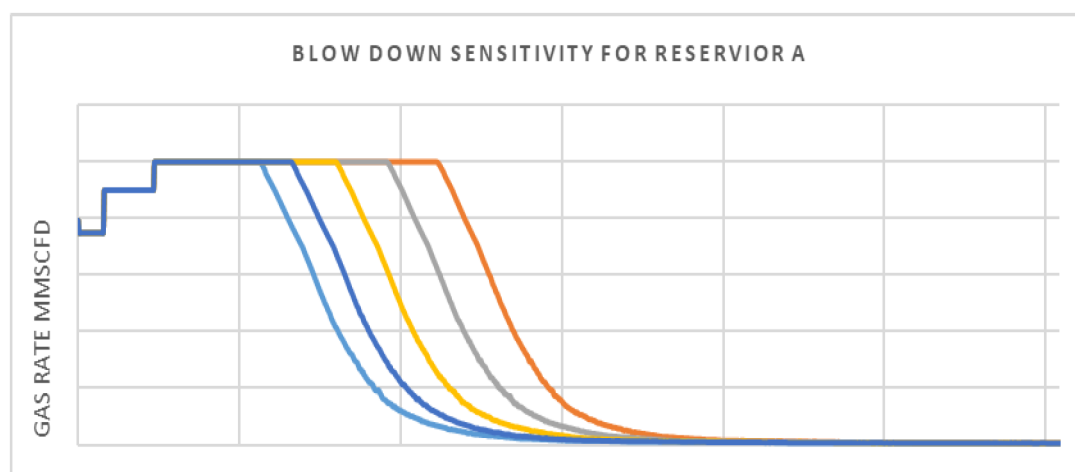


Figure 5—Sensitivity to Blowdown Timing

Compression delivered substantial benefits in depletion systems. With no injection support, WHP declined rapidly, and compression extended economic life by 8–10 years. The timing of compressor deployment was key; installing too early reduced gas rate acceleration, while delaying beyond 1000 psi reservoir pressure reduced available drawdown. Inline compression proved suitable for individual high permeability wells, while centralized compression offered system-wide efficiency in tight zones.

Infill drilling success depended heavily on pressure support and remaining resource mapping. When executed in high saturation regions with active GI, new wells delivered 10–15 MMscfd initial rates with low drawdown. In depletion systems, infill productivity was lower and required compression to sustain flow. Combining infill with other levers (e.g., THP reduction and compression) yielded the highest incremental value.

Discussion

Quantitative comparison across scenarios showed that GI to GP added an average of 6.5 years of plateau in the top two reservoirs, while adding only 1.1 years in the lowest. THP reduction at 200 psia increased cumulative gas production by an average of 8.7%, while infill drilling contributed an additional 5.2% in recoverable gas when strategically placed. Blowdown initiation timing, when delayed to a specific pressure threshold (typically ~1200 psi), protected condensate yields by ~4% on average. These findings emphasize the value of scenario-based lever screening using validated models.

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In addition to top performing reservoirs, two assets reservoir C and E showed limited responsiveness to several levers. In reservoir C, a highly laminated formation with low vertical permeability, GI to GP conversion delivered only 1.2 years of plateau extension. Tracer simulation indicated poor vertical sweep and limited injector connectivity, explaining the marginal benefit. Condensate recovery improved by only 0.8%, and skin evolution data suggested that conversion induced drawdown created near wellbore liquid loading.

The integrated simulation-based approach confirms that redevelopment planning must be customized to each reservoir's maturity, fluid properties, and infrastructure constraints. Rather than adopting a uniform strategy, development levels must be sequenced and selected based on reservoir specific response. GI to GP conversion is most effective in high pressure reservoirs with strong historical injections and connected pore volume. These systems showed an uplift in plateau life and condensate recovery due to mobilization of previously trapped fluids near injectors.

THP reduction is effective across all systems but delivers diminishing returns beyond 200 psia. The simulation also showed that extreme THP reduction can result in higher liquid dropout near the wellbore, which can be mitigated with wellbore cleanout or surfactant injection. Blowdown scheduling emerges as a critical optimization lever, particularly in reservoirs with marginal condensate yields. The timing must account for surface facility flexibility and market price signals.

Compression remains a high impact lever in depletion assets. However, careful timing and network modeling are essential. Deploying compression too early reduces system drawdown, while deploying too late leaves gas unrecovered. Combining compression with targeted infill enables producers to access bypassed gas zones and maintain rates post plateau. Finally, infill drilling on its own offers moderate gains but when combined with pressure support and surface debottlenecking, it becomes a powerful tool to flatten decline and accelerate recovery.

Overall, the most effective redevelopment strategies involve combining at least three levers per reservoir in a phased, simulation driven plan. Sensitivity analysis showed that dual lever combinations consistently outperformed individual applications in both gas rate uplift and condensate recovery.

Conclusion

Going forward, this methodology can be extended to unconventional gas condensate systems, or to assets undergoing enhanced condensate recovery techniques such as hydrocarbon gas recycling. Multi reservoir models incorporating cross asset swing behavior are also recommended for system wide optimization.

This study underscores the need for a simulation driven lever integrated planning approach to redevelopment of mature gas condensate reservoirs. In applying this method, the following technical conclusions were drawn:

- Model resolution and boundary assumptions significantly influence lever performance prediction. High fidelity simulation is essential.
- Performance gains are not uniform across reservoirs. Understanding dynamic connectivity, pressure communication, and local saturation history is key to optimizing lever sequencing.
- Underperforming cases offer learning opportunities. Poor GI to GP results can often be traced to inadequate sweep or limited mobility, while ineffective infill cases highlight the risk of compartmentalization.
- Integrated simulation combining subsurface and surface networks enables better prediction of bottlenecks and compression timing.
- Quantifying the marginal gain per lever enables phased implementation with minimal CAPEX exposure.

This study demonstrates that a simulation driven, multi lever redevelopment strategy can significantly extend plateau and improve recovery in mature gas condensate reservoirs. The following conclusions were drawn:

1. GI to GP conversion provides high value when applied to connected, pressure supported reservoirs, yielding plateau extensions of up to 11 years.
2. THP reduction to 200 psia offers optimal uplift while maintaining operational safety. Further reduction adds limited value and may introduce mechanical risks.
3. Blowdown must be optimized per reservoir and system constraints. Timing materially impacts condensate yield and surface handling.
4. Compression is critical in depletion systems and must be carefully staged. Its synergy with infill drilling unlocks late life productivity.
5. Infill drilling alone provides moderate benefit, but in combination with other levers, it accelerates recovery and production stabilization.
6. Simulation enables accurate performance prediction and lever selection, ensuring efficient investment and execution planning.

The approach is transferable to similar gas condensate fields seeking to optimize value in late life production.

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